

# Life Processes

Nutrition in plants and animals, respiration and breathing, transport of blood and water, and excretion in humans and plants, with diagrams and practice questions.

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## Introduction

All living things need energy to stay alive. They need to grow, repair and reproduce. For this they need food, oxygen and removal of wastes. These basic functions are called life processes. In this chapter we study four main life processes: nutrition, respiration, transport and excretion. These processes work together to keep the body alive.

## Nutrition

### Nutrition

The process by which an organism takes in food and uses it for energy, growth and repair.

There are two main modes of nutrition:

- **Autotrophic nutrition.** The organism makes its own food from simple inorganic substances. Example: green plants, some bacteria.
- **Heterotrophic nutrition.** The organism cannot make its own food. It depends on other organisms for food. Example: animals, fungi, many bacteria.

## Autotrophic Nutrition and Photosynthesis

Green plants make food by photosynthesis. They use carbon dioxide from air, water from soil, and sunlight. Chlorophyll present in chloroplasts traps sunlight.

### Photosynthesis

The process by which green plants make food (glucose) from carbon dioxide and water in the presence of sunlight and chlorophyll, and release oxygen.



### Steps and events in photosynthesis:

- Absorption of light energy by chlorophyll.
- Conversion of light energy to chemical energy and splitting of water into hydrogen and oxygen. This is called photolysis of water.
- Reduction of carbon dioxide to form carbohydrates like glucose.

### Raw materials:

- Carbon dioxide enters through stomata. Stomata are tiny pores on leaves. Each stoma is surrounded by two guard cells which control its opening and closing.
- Water is taken from soil by roots and carried to leaves through xylem.
- Sunlight is trapped by chlorophyll.

Excess glucose is stored as starch in plants. The oxygen released is used for respiration.

## Heterotrophic Nutrition

Heterotrophic nutrition has three types:

- **Holozoic:** Whole food is taken and digested inside the body. Example: humans, amoeba.
- **Saprophytic:** Food is taken from dead and decaying matter. The organism releases enzymes outside to digest the food. Example: fungi, bread mould.
- **Parasitic:** The organism lives inside or on another living organism and takes food from it. Example: cuscuta, tapeworm, leech.

**Nutrition in Amoeba.** Amoeba is a single celled organism. It takes food by the process called endocytosis. It extends pseudopodia around the food particle and forms a food vacuole. Enzymes digest the food inside the vacuole. Undigested food is thrown out. This is holozoic nutrition.

**Nutrition in Paramecium.** Paramecium takes food at a fixed spot. Cilia move food to the mouth region.

## Nutrition in Human Beings

Human beings have a complete alimentary canal from mouth to anus. Digestion is the breaking of large insoluble food into small soluble forms by enzymes.

### Parts of the human alimentary canal:

- **Mouth:** Food is crushed by teeth. Salivary glands release saliva. Saliva has salivary amylase which breaks starch into sugar. The tongue mixes food with saliva.
- **Oesophagus:** Food moves down by peristalsis. Peristalsis is the rhythmic contraction of muscles of the canal.
- **Stomach:** Gastric glands release HCl, pepsin and mucus. HCl makes the medium acidic and kills germs. Pepsin digests proteins. Mucus protects the stomach lining.
- **Small intestine:** It is the main site of digestion. It receives bile from the liver and pancreatic juice from the pancreas. Bile breaks large fat globules into small ones. This is emulsification. Pancreatic juice has trypsin for proteins and lipase for fats. Intestinal juice completes digestion. Glucose, amino acids, fatty acids and glycerol are absorbed through villi. Villi are finger like projections in the small intestine. They increase surface area for absorption and have blood vessels.
- **Large intestine:** It absorbs water from undigested food. The remaining waste is removed through the anus. This is egestion.

**Role of accessory glands:** Liver is the largest gland. It makes bile which is stored in the gall bladder. Pancreas makes pancreatic juice.

## Respiration

### Respiration

The process of breaking down food (glucose) to release energy in the form of ATP.

Respiration is different from breathing. Breathing is the physical process of taking in oxygen and giving out carbon dioxide. Respiration is the chemical process inside cells.

### **Types of respiration:**

- **Aerobic respiration:** Takes place in the presence of oxygen. Glucose is broken down to carbon dioxide and water, releasing much more energy than anaerobic respiration.
- **Anaerobic respiration:** Takes place in absence of oxygen. Less energy is released.

Aerobic: Glucose + Oxygen → Carbon dioxide + Water + Energy

Anaerobic in yeast: Glucose → Ethanol + Carbon dioxide + Energy

Anaerobic in muscle cells: Glucose → Lactic acid + Energy

When we do heavy exercise, our muscle cells respire anaerobically. Lactic acid builds up and causes cramps. A hot bath helps by improving blood flow.

### **Human respiratory system:**

- Air enters through nostrils, then passes through nasal passage, pharynx, larynx, trachea, bronchi, bronchioles and finally alveoli.
- Alveoli are tiny air sacs in the lungs. They have thin walls and a large surface area. They are surrounded by blood capillaries. Exchange of O<sub>2</sub> and CO<sub>2</sub> takes place here.
- The trachea has rings of cartilage to prevent it from collapsing.
- In humans, breathing rate is about 12 to 20 times per minute. During exercise it increases.

**Mechanism of breathing:** When we inhale, the ribs move up and the diaphragm flattens. Chest cavity becomes larger. Air rushes in. When we exhale, the ribs move down and the diaphragm becomes dome shaped. Air is pushed out.

**Respiration in plants:** Plants exchange gases through stomata and lenticels. They have no special respiratory system. At night they take O<sub>2</sub> and release CO<sub>2</sub>. During the day CO<sub>2</sub> produced is used for photosynthesis.

# Transport in Human Beings

Transport is the movement of food, oxygen, hormones, wastes and other materials inside the body.

**Blood.** Blood is a fluid connective tissue. It has four main parts:

- **Plasma:** Liquid part. Carries dissolved substances like food, CO<sub>2</sub> and wastes.
- **Red Blood Cells (RBC):** Contain haemoglobin. Haemoglobin binds with oxygen and carries it. RBCs have no nucleus in mammals.
- **White Blood Cells (WBC):** Fight germs and protect the body.
- **Platelets:** Help in blood clotting at the site of injury.

**Blood vessels:**

- **Arteries:** Carry blood away from the heart. They have thick elastic walls. Blood pressure is high inside them. They carry oxygenated blood except the pulmonary artery.
- **Veins:** Carry blood towards the heart. They have thin walls and valves to prevent backflow. They carry deoxygenated blood except the pulmonary vein.
- **Capillaries:** Very thin vessels where exchange of materials takes place between blood and cells. Walls are only one cell thick.

**Lymphatic system.** Lymph is a light yellow fluid that leaks from capillaries into spaces between cells. It flows in lymph vessels and collects in lymph nodes. Lymph carries digested fat from the intestine and drains excess fluid back to blood. It also helps in fighting infection.

## Heart and Blood Circulation

The human heart is a muscular organ about the size of a fist. It is located slightly to the left side of the chest. It has four chambers: two upper atria and two lower ventricles. The left and right sides are separated by a septum. This prevents mixing of oxygenated and deoxygenated blood.

**Working of the heart:**

- Deoxygenated blood from the body comes to the right atrium through vena cava. It goes to the right ventricle and is pumped to the lungs through the pulmonary artery.

- In the lungs, blood gets oxygenated and returns to the left atrium through the pulmonary vein.
- From the left atrium it goes to the left ventricle. The left ventricle pumps oxygenated blood to the whole body through the aorta.

This is called double circulation because blood passes through the heart twice in one complete cycle.

- **Pulmonary circulation:** Right ventricle to lungs to left atrium.
- **Systemic circulation:** Left ventricle to body organs to right atrium.

Double circulation is needed because humans are warm blooded. They need constant oxygen supply to maintain body temperature. No mixing of oxygenated and deoxygenated blood gives more efficient oxygen supply.

**Heartbeat:** The heart beats about 70 to 72 times per minute. When the heart contracts, it is systole. When it relaxes, it is diastole. Blood pressure is measured as systolic pressure over diastolic pressure. Normal value is 120/80 mm of Hg.

#### Remember

Arteries carry blood away from the heart. Veins carry blood to the heart. Capillaries connect arteries and veins. Pulmonary artery is the only artery with deoxygenated blood. Pulmonary vein is the only vein with oxygenated blood.

## Transport in Plants

Plants need to transport water, minerals and food.

- **Xylem** carries water and minerals from roots to leaves. Movement is by transpiration pull. Transpiration is loss of water vapour from leaves through stomata. It creates suction that pulls water up.
- **Phloem** carries food made in leaves to all parts of the plant. This movement is called translocation. It needs energy from ATP. Food can move up or down, unlike water which moves only up.

Transpiration also helps to cool the plant and keeps the plant firm.

# Excretion

## Excretion

The process of removing harmful metabolic wastes, mainly nitrogenous wastes, from the body.

In humans, the main excretory system consists of two kidneys, two ureters, one urinary bladder and one urethra.

- **Kidneys:** Bean shaped organs located on either side of the backbone. They filter blood and form urine. They also control water balance.
- **Ureters:** Tubes that carry urine from kidneys to the bladder.
- **Urinary bladder:** Muscular bag that stores urine till it is released.
- **Urethra:** Tube through which urine leaves the body.

The kidney also controls the amount of water and salts in the body. This is called osmoregulation.

## Structure and Working of Nephron

The nephron is the basic filtering unit of the kidney. Each kidney has about one million nephrons.

### Parts of a nephron:

- **Bowman's capsule:** Cup shaped structure that surrounds the glomerulus.
- **Glomerulus:** A cluster of capillaries where blood is filtered under pressure.
- **Tubule:** Long coiled tube with proximal tubule, loop of Henle, and distal tubule.
- **Collecting duct:** Collects urine from many nephrons and carries it to the ureter.

### Steps of urine formation:

- **1. Filtration:** Blood enters the glomerulus. Water, glucose, salts, urea and other wastes are filtered into Bowman's capsule. This filtrate is called glomerular filtrate. Blood cells and proteins are not filtered.
- **2. Selective reabsorption:** Useful substances like glucose, amino acids, salts and most of the water are reabsorbed from the tubule back into the blood.
- **3. Tubular secretion:** Extra wastes and H<sup>+</sup> ions are secreted into the tubule.

- **4. Collection:** The remaining fluid is urine. It passes to the collecting duct and then to the ureter.

Urine contains water, urea and other dissolved wastes. Its volume and composition vary with water intake, diet, activity and environmental conditions.

**Artificial kidney (Haemodialysis).** When kidneys fail, blood is filtered by a machine. Blood is taken from an artery, passed through tubes with semipermeable membrane in dialysing fluid, and then returned to a vein. This removes nitrogenous wastes.

**Excretion in plants:** Plants do not have a special excretory system. They store wastes in different ways. Oxygen from photosynthesis is released through stomata. Excess water is lost by transpiration. Some wastes are stored in dead tissues like bark, in leaves that fall off, or as gums and resins. Plants also release wastes into the soil.

## Summary

- Life processes are nutrition, respiration, transport and excretion.
- Autotrophic nutrition is making food by photosynthesis. Heterotrophic nutrition depends on other organisms. It can be holozoic, saprophytic or parasitic.
- Human digestion breaks food into small soluble parts. Villi in the small intestine absorb digested food.
- Respiration releases energy from glucose. Aerobic respiration uses oxygen and produces carbon dioxide and water; anaerobic respiration releases less energy.
- Alveoli in lungs are the site of gas exchange. Breathing is by movement of ribs and diaphragm.
- Blood has plasma, RBC, WBC and platelets. Arteries, veins and capillaries are the three types of blood vessels.
- The heart has four chambers and shows double circulation. Normal blood pressure is 120/80 mm Hg.
- Xylem carries water up and phloem carries food in both directions.
- Kidneys filter blood and form urine. The nephron is the filtering unit. Urine formation has filtration, reabsorption and secretion.
- Plants store wastes as resins, gums or in falling leaves. They have no kidney like system.

## Practice Questions

1. What is the difference between autotrophic and heterotrophic nutrition? Give one example of each.  

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2. Write the equation for photosynthesis and explain where each raw material comes from.  

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3. Explain the process of digestion in the small intestine. What is the role of villi?  

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4. Differentiate between aerobic and anaerobic respiration. Write their equations.  

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5. Describe the structure of the human heart with a labelled diagram. Why is double circulation needed?  

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6. What are arteries, veins and capillaries? How are they different from each other?  

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7. Draw a labelled diagram of the nephron and explain the steps of urine formation.  

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8. How do xylem and phloem help in transport in plants?